

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	Bheeshma L	Reg. No.:	192511332
Date:	Aug 19	Duration:	60 minutes

Code of Conduct

I Bheeshma L / 192511332 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bheeshma L

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15 %)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25 %)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied

Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

CO4 AT-1 TEST CASE DESIGN DOCUMENT

Intelligent Water Quality Assessment and Contamination Detection Platform

1. Introduction

The **Intelligent Water Quality Assessment and Contamination Detection Platform** is a software-based system designed to monitor, analyze, and assess the quality of water using sensor data and intelligent analysis techniques. The platform collects water-quality parameters such as pH, temperature, turbidity, dissolved oxygen, electrical conductivity, and other relevant parameters.

The system analyzes the collected data against predefined acceptable water-quality ranges and identifies possible contamination. When abnormal values or contamination patterns are detected, the platform generates alerts and provides appropriate information to authorized users.

The purpose of this test case design document is to verify that the proposed platform satisfies its functional and non-functional requirements. The testing process covers normal, boundary, invalid, exceptional, and security-related conditions.

2. Objectives of Testing

The major objectives of testing are:

1. To verify that water-quality sensor data is correctly collected by the system.
2. To verify that valid sensor readings are correctly processed and stored.
3. To verify that invalid or incomplete sensor data is rejected or handled appropriately.
4. To verify that the system correctly identifies water-quality conditions.
5. To verify that contamination is detected when abnormal parameter values are observed.
6. To verify that alerts are generated when contamination or unsafe conditions are detected.
7. To verify that authorized users can view water-quality reports and dashboards.
8. To verify that unauthorized users cannot access protected information.
9. To verify that the system handles sensor failures and communication failures correctly.
10. To verify system performance, reliability, usability, and security.

3. Scope of Testing

The testing scope includes the following major components:

- User registration and login
- User authentication and authorization
- Sensor data collection
- Water-quality parameter validation
- Data processing and analysis
- Water-quality classification
- Contamination detection
- Alert and notification generation
- Dashboard and visualization
- Historical data storage
- Report generation
- Sensor/device management
- Error handling
- Security
- Performance and reliability

The testing does not include physical laboratory validation of water samples unless such functionality is specifically integrated into the platform.

4. Functional Requirements

FR-01: User Registration

The system shall allow new users to register by providing required information such as name, email address, username, and password.

FR-02: User Login

The system shall authenticate registered users using valid credentials.

FR-03: Role-Based Access

The system shall provide different access privileges for administrators, monitoring staff, and normal users.

FR-04: Sensor Data Collection

The system shall receive water-quality readings from connected sensors or data sources.

FR-05: Parameter Validation

The system shall validate sensor readings before processing them.

FR-06: Water Quality Assessment

The system shall analyze water-quality parameters and determine whether the water is within acceptable limits.

FR-07: Contamination Detection

The system shall identify potentially contaminated water based on abnormal or dangerous parameter values.

FR-08: Alert Generation

The system shall generate an alert when a critical water-quality condition or possible contamination is detected.

FR-09: Dashboard

The system shall display current water-quality information through a user-friendly dashboard.

FR-10: Historical Data

The system shall store historical readings and allow authorized users to review previous water-quality conditions.

FR-11: Report Generation

The system shall generate water-quality assessment reports.

FR-12: Sensor Failure Handling

The system shall identify missing, invalid, or unavailable sensor readings and notify the appropriate user when necessary.

FR-13: Data Filtering

The system shall allow users to filter water-quality records according to parameters such as date, location, sensor, or contamination status.

FR-14: Notifications

The system shall notify authorized users when critical contamination conditions are detected.

FR-15: Logout

The system shall allow users to securely log out of the platform.

5. Non-Functional Requirements

NFR-01: Performance

The system should process incoming sensor readings within an acceptable response time.

NFR-02: Reliability

The platform should continue operating correctly when individual sensor readings are missing or temporarily unavailable.

NFR-03: Availability

The platform should be available to authorized users whenever monitoring services are required.

NFR-04: Security

User credentials, sensor information, and water-quality data should be protected against unauthorized access.

NFR-05: Usability

The dashboard and reports should be simple and understandable for users with basic technical knowledge.

NFR-06: Scalability

The system should support additional sensors and monitoring locations without major performance degradation.

NFR-07: Maintainability

The system should be designed so that software components and water-quality thresholds can be updated without affecting unrelated modules.

NFR-08: Accuracy

The system should correctly process sensor readings and classify water quality according to configured standards and thresholds.

NFR-09: Compatibility

The platform should work correctly on commonly used web browsers and supported devices.

NFR-10: Data Integrity

Water-quality readings should not be unintentionally modified, duplicated, or lost during storage and processing.

6. Test Design Techniques

The following test design techniques will be used:

6.1 Equivalence Partitioning

Input values will be divided into valid and invalid groups.

For example:

- Valid pH values
- Below acceptable pH values
- Above acceptable pH values
- Non-numeric pH values
- Empty pH values

6.2 Boundary Value Analysis

Values at the minimum and maximum acceptable limits will be tested.

For example, if the configured acceptable pH range is 6.5–8.5:

- 6.4 – below boundary
- 6.5 – minimum boundary
- 6.6 – just above minimum
- 8.4 – just below maximum
- 8.5 – maximum boundary
- 8.6 – above boundary

6.3 Decision Table Testing

Different combinations of water-quality conditions will be tested to determine whether the system correctly classifies water and generates alerts.

6.4 State Transition Testing

The system will be tested through states such as:

Normal → Warning → Critical → Resolved

6.5 Error Guessing

Potential problems such as missing sensor data, network failure, duplicate readings, invalid credentials, and unexpected input will be tested.

7. Test Scenarios

Scenario ID	Test Scenario
TS-01	Register a new user
TS-02	Login using valid credentials
TS-03	Login using invalid credentials
TS-04	Receive valid sensor readings
TS-05	Process missing sensor data
TS-06	Validate pH readings
TS-07	Validate turbidity readings
TS-08	Validate temperature readings
TS-09	Detect abnormal water-quality readings
TS-10	Detect possible contamination
TS-11	Generate contamination alert
TS-12	Display real-time dashboard

Scenario ID	Test Scenario
TS-13	Store historical readings
TS-14	Search and filter historical records
TS-15	Generate water-quality report
TS-16	Handle sensor communication failure
TS-17	Prevent unauthorized access
TS-18	Logout from the system
TS-19	Handle duplicate sensor readings
TS-20	Test system performance under high data volume

8. Detailed Test Cases

TC-001: User Registration with Valid Data

Test Scenario: Register a new user.

Precondition: Registration page is available.

Test Steps:

1. Open the registration page.
2. Enter a valid name.
3. Enter a valid email address.
4. Enter a unique username.
5. Enter a valid password.
6. Click the Register button.

Test Data:

Name: Test User

Email: testuser@example.com

Username: wateruser01

Password: Water@123

Expected Result:

The system successfully creates the user account and displays a registration-success message.

Actual Result:

To be recorded during execution.

Status: Pass/Fail

TC-002: User Registration with Missing Required Field

Test Scenario: Register without entering an email address.

Test Steps:

1. Open registration.
2. Enter all required information except email.
3. Click Register.

Expected Result:

The system should reject the registration and display an appropriate validation message.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-003: Login with Valid Credentials

Test Scenario: Authenticate a registered user.

Test Steps:

1. Open the login page.
2. Enter a registered username.
3. Enter the correct password.
4. Click Login.

Expected Result:

The user is authenticated and redirected to the appropriate dashboard.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-004: Login with Invalid Password

Test Scenario: Attempt login using an incorrect password.

Test Data:

Username: wateruser01

Password: WrongPassword

Expected Result:

The system denies access and displays an appropriate error message without exposing sensitive information.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-005: Valid pH Reading

Test Scenario: Process a normal pH value.

Test Data:

pH = 7.0

Test Steps:

1. Send a valid sensor reading with pH 7.0.
2. Submit the reading for analysis.
3. View the assessment result.

Expected Result:

The system accepts the reading and classifies the pH condition as normal according to the configured water-quality standard.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-006: pH Lower Boundary

Test Scenario: Test pH at the configured minimum acceptable boundary.

Test Data:

pH = 6.5

Expected Result:

The system accepts the value and classifies it according to the configured threshold rules.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-007: pH Below Acceptable Range

Test Scenario: Detect acidic water condition.

Test Data:

pH = 5.5

Expected Result:

The system identifies the abnormal pH condition and generates a warning or contamination alert according to configured rules.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-008: pH Above Acceptable Range

Test Scenario: Detect highly alkaline water.

Test Data:

pH = 9.5

Expected Result:

The system identifies the abnormal reading and displays an appropriate warning or critical alert.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-009: Invalid pH Input

Test Scenario: Send non-numeric pH data.

Test Data:
pH = "ABC"

Expected Result:
The system rejects the invalid reading and records an appropriate validation/error message.

Actual Result:
To be recorded.

Status: Pass/Fail

TC-010: Missing Sensor Reading

Test Scenario: Process incomplete sensor data.

Test Data:
pH = 7.2
Temperature = 26°C
Turbidity = NULL

Expected Result:
The system identifies the missing value and prevents an incorrect complete assessment. The missing parameter should be clearly indicated.

Actual Result:
To be recorded.

Status: Pass/Fail

TC-011: High Turbidity Detection

Test Scenario: Detect unusually high turbidity.

Test Data:
Turbidity = 100 NTU

Expected Result:

The system identifies the turbidity value as abnormal according to the configured threshold and generates an appropriate warning.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-012: Normal Temperature Reading

Test Scenario: Process a normal temperature value.

Test Data:

Temperature = 25°C

Expected Result:

The system accepts the reading and displays the corresponding water-quality condition.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-013: Multiple Abnormal Parameters

Test Scenario: Detect contamination using multiple abnormal readings.

Test Data:

- pH = 4.8
- Turbidity = 150 NTU
- Dissolved Oxygen = Very Low
- Temperature = 35°C

Expected Result:

The system identifies the combined abnormal condition as potentially unsafe and generates a high-priority contamination alert.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-014: Contamination Alert Generation

Test Scenario: Verify automatic alert generation.

Test Steps:

1. Send sensor data containing a critical abnormal value.
2. Allow the system to analyze the data.
3. Check the alert/notification section.

Expected Result:

A contamination alert is generated containing relevant information such as location, affected parameter, severity, timestamp, and recommended action.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-015: Dashboard Display

Test Scenario: Verify real-time water-quality dashboard.

Test Steps:

1. Login as an authorized user.
2. Open the monitoring dashboard.
3. Observe the displayed water-quality parameters.

Expected Result:

The dashboard displays current sensor values, water-quality status, alerts, timestamps, and relevant visual indicators correctly.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-016: Historical Data Retrieval

Test Scenario: View previously stored water-quality readings.

Test Steps:

1. Login as an authorized user.
2. Open historical data.
3. Select a date range.
4. View the records.

Expected Result:

The system displays stored readings for the selected period accurately.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-017: Data Filtering

Test Scenario: Filter records by location and date.

Test Steps:

1. Open historical records.
2. Select a monitoring location.
3. Select a date range.
4. Click Search/Filter.

Expected Result:

Only records matching the selected criteria are displayed.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-018: Report Generation

Test Scenario: Generate a water-quality report.

Test Steps:

1. Select the required monitoring location.
2. Select a date range.
3. Select Generate Report.
4. Download or view the generated report.

Expected Result:

The system generates a report containing relevant readings, water-quality classifications, detected abnormal conditions, alerts, and timestamps.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-019: Sensor Communication Failure

Test Scenario: Simulate loss of communication between a sensor and the platform.

Test Steps:

1. Disconnect or disable a sensor.
2. Wait for the configured communication timeout.
3. Check the monitoring dashboard.

Expected Result:

The system identifies the sensor as unavailable and displays an appropriate warning. It should not interpret missing data as normal water quality.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-020: Duplicate Sensor Data

Test Scenario: Send the same sensor reading multiple times.

Test Data:

Same sensor ID, timestamp, and readings submitted twice.

Expected Result:

The system should detect or appropriately handle duplicate records and prevent incorrect duplication in historical analysis.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-021: Unauthorized Dashboard Access

Test Scenario: Attempt to access the dashboard without authentication.

Test Steps:

1. Log out of the system.
2. Directly enter the dashboard URL.
3. Attempt to access the dashboard.

Expected Result:

The system denies access and redirects the user to the login page.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-022: Role-Based Access Control

Test Scenario: Verify restricted administrative functionality.

Test Steps:

1. Login as a normal user.
2. Attempt to access administrative functions such as sensor configuration or user management.

Expected Result:

The system prevents unauthorized users from accessing administrator-only functionality.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-023: Logout

Test Scenario: Securely log out of the platform.

Test Steps:

1. Login successfully.
2. Navigate to the dashboard.
3. Click Logout.
4. Attempt to access the dashboard using the browser back button.

Expected Result:

The user is logged out and protected pages cannot be accessed without authentication.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-024: High Data Volume Performance

Test Scenario: Test system behavior when a large number of sensor readings are received.

Test Steps:

1. Simulate multiple sensors sending readings simultaneously.
2. Monitor system response time.
3. Verify data storage and dashboard behavior.

Expected Result:

The system processes the incoming data without crashing, losing records, or causing unacceptable delays.

Actual Result:

To be recorded.

Status: Pass/Fail

TC-025: Recovery After Temporary Network Failure

Test Scenario: Verify system recovery after network interruption.

Test Steps:

- 1. Establish normal sensor communication.
- 2. Temporarily interrupt the network.
- 3. Restore the network.
- 4. Observe sensor communication and stored data.

Expected Result:

The system recovers gracefully after connectivity is restored and processes new readings correctly. Where supported, temporarily buffered readings should be handled according to the system design.

Actual Result:

To be recorded.

Status: Pass/Fail

9. Sample Test Data

Parameter	Normal Example	Boundary/Warning Example	Invalid Example
pH	7.0	6.5 / 8.5	ABC
Temperature	25°C	Configured upper/lower threshold	XYZ
Turbidity	Low/acceptable value	High NTU value	-ABC
Dissolved Oxygen	Acceptable value	Low threshold value	NULL
Conductivity	Acceptable value	High threshold value	Text
Sensor ID	SENSOR001	SENSOR999	Empty
Timestamp	Valid timestamp	Future timestamp	Invalid date
Location	Valid location	Unregistered location	Empty

Note: Exact water-quality thresholds should be configured according to the standard selected for your project and the requirements of your implementation.

10. Decision Table for Contamination Detection

Condition	pH	Turbidity	Dissolved Oxygen	Expected System Decision
D1	Normal	Normal	Normal	Safe/Normal
D2	Abnormal	Normal	Normal	Warning
D3	Normal	Abnormal	Normal	Warning
D4	Normal	Normal	Low	Warning
D5	Abnormal	Abnormal	Normal	High Risk
D6	Abnormal	Abnormal	Low	Critical/Contamination Alert
D7	Missing	Normal	Normal	Incomplete Data
D8	Invalid	Normal	Normal	Reject Input

11. State Transition Testing

The water-quality monitoring system can be represented using the following states:

Normal → Warning → Critical → Resolved

Normal State

All monitored parameters are within configured acceptable ranges.

Warning State

One or more parameters show abnormal values that require attention.

Critical State

Critical parameters or multiple abnormal conditions indicate a potentially serious contamination event.

Resolved State

The abnormal condition has returned to an acceptable level or has been acknowledged and resolved by an authorized user.

The system should correctly transition between these states based on incoming sensor readings.

12. Non-Functional Test Cases

NFT-001: Response Time

Objective: Verify dashboard response time.

Test: Open the dashboard while valid sensor data is available.

Expected Result: The dashboard should load within the performance target defined by the project.

NFT-002: Security

Objective: Verify protection of user accounts.

Test: Attempt to access protected resources without authentication.

Expected Result: Unauthorized access is denied.

NFT-003: Usability

Objective: Verify dashboard usability.

Test: Ask representative users to locate current water-quality status, alerts, and historical information.

Expected Result: Users should be able to perform the required tasks without confusion.

NFT-004: Reliability

Objective: Verify system behavior during sensor failure.

Test: Disconnect one sensor while other sensors remain operational.

Expected Result: The affected sensor is reported as unavailable while other monitoring functions continue operating.

NFT-005: Scalability

Objective: Determine whether the platform can support an increasing number of sensors.

Test: Gradually increase the number of simulated sensor data streams.

Expected Result: The system should continue processing data within the defined performance requirements.

13. Exception and Error Handling Tests

The following exceptional situations should be tested:

1. Internet/network connection failure.
2. Sensor disconnected.
3. Sensor sends incomplete data.
4. Sensor sends corrupted data.
5. Database temporarily unavailable.
6. Duplicate sensor readings.
7. Invalid timestamp.
8. Unregistered sensor sends data.
9. Unauthorized user attempts administrative access.
10. User session expires.
11. Report generation fails.
12. Notification service becomes unavailable.
13. Large volume of sensor data is received.
14. Unexpected application/server failure.

For each exception, the system should display an understandable error message, preserve data integrity, and recover without causing incorrect water-quality assessments.

14. Requirement Traceability Matrix

Requirement	Related Test Cases
FR-01 User Registration	TC-001, TC-002
FR-02 User Login	TC-003, TC-004
FR-03 Role-Based Access	TC-022

Requirement	Related Test Cases
FR-04 Sensor Data Collection	TC-005, TC-012
FR-05 Parameter Validation	TC-006, TC-007, TC-009, TC-010
FR-06 Water Quality Assessment	TC-005, TC-006, TC-007, TC-008
FR-07 Contamination Detection	TC-013
FR-08 Alert Generation	TC-014
FR-09 Dashboard	TC-015
FR-10 Historical Data	TC-016
FR-11 Report Generation	TC-018
FR-12 Sensor Failure Handling	TC-019
FR-13 Data Filtering	TC-017
FR-14 Notifications	TC-014
FR-15 Logout	TC-023
NFR-01 Performance	TC-024
NFR-02 Reliability	TC-019, TC-025
NFR-04 Security	TC-021, TC-022, TC-023
NFR-05 Usability	NFT-003
NFR-06 Scalability	NFT-005
NFR-10 Data Integrity	TC-020, TC-025

15. Test Coverage

The proposed test suite provides coverage for:

Functional Coverage

- User authentication
- Sensor data collection
- Data validation
- Water-quality assessment
- Contamination detection
- Alert generation
- Dashboard
- Historical records
- Reporting
- Sensor failure
- Access control

Input Coverage

- Valid inputs
- Invalid inputs
- Empty inputs
- Boundary values
- Abnormal values
- Duplicate values
- Missing values

Exception Coverage

- Network failure
- Sensor failure
- Invalid data
- Database/service failure
- Unauthorized access
- Session expiration
- High-volume data

Non-Functional Coverage

- Performance
- Security
- Reliability
- Usability
- Scalability
- Data integrity
- Compatibility

16. Entry Criteria

Testing can begin when:

1. The required software build is available.
 2. Major system modules are implemented.
 3. Test environment is configured.
 4. Test database is available.
 5. Required sensors or simulated sensor data are available.
 6. Water-quality thresholds have been configured.
 7. Test accounts are available.
 8. Test data has been prepared.
-

17. Exit Criteria

Testing can be considered complete when:

1. All planned test cases have been executed.
 2. All critical test cases have passed.
 3. No unresolved critical defects remain.
 4. Major functional requirements have adequate test coverage.
 5. Security and access-control tests have been completed.
 6. Performance testing has been completed.
 7. Test results have been documented.
 8. Known limitations and remaining defects have been recorded.
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18. Defect Reporting

Any failed test should result in a defect report containing:

- Defect ID
- Test Case ID
- Defect title
- Description
- Steps to reproduce
- Test data
- Expected result
- Actual result
- Severity
- Priority
- Screenshot/evidence
- Assigned developer
- Defect status

Example

Defect ID: BUG-001

Test Case: TC-014

Title: Contamination alert not generated

Severity: Critical

Description: The system detects a critical combination of abnormal water-quality parameters but does not generate an alert.

Expected Result: A critical contamination notification should be generated.

Actual Result: No notification is generated.

Status: Open

19. Test Execution Summary

Category	Planned Tests	Passed	Failed	Blocked
Functional Testing	25	To be filled	To be filled	To be filled
Security Testing	3	To be filled	To be filled	To be filled
Performance Testing	1	To be filled	To be filled	To be filled
Usability Testing	1	To be filled	To be filled	To be filled
Reliability Testing	2	To be filled	To be filled	To be filled

20. Conclusion

The **Intelligent Water Quality Assessment and Contamination Detection Platform** requires comprehensive testing because incorrect water-quality information may result in inappropriate decisions. The proposed test cases cover both normal and abnormal system behavior, including valid inputs, boundary values, invalid inputs, sensor failures, contamination conditions, security violations, and high-volume data processing.

Equivalence partitioning, boundary value analysis, decision table testing, state transition testing, and error guessing techniques are applied to achieve effective test coverage.

Successful completion of these tests will provide evidence that the platform can reliably collect and analyze water-quality information, detect potentially contaminated conditions, generate appropriate alerts, protect user information, and provide useful monitoring and reporting functionality.